

Literature Survey (Step 9): PID-Based Steering Control for Lane Keeping in Autonomous Vehicles

Maintaining an autonomous vehicle within its intended lane or track requires robust lateral control under uncertainty in tire–road interaction, vehicle parameters, sensing noise, actuator limits, and computation/communication delay. Among lateral control methods, the proportional–integral–derivative (PID) family remains widely deployed due to its transparency, low computational burden, and ease of verification. However, lane-keeping exposes PID controllers to challenges such as parameter variation with speed, nonlinearities (tire saturation, steering mechanism dynamics), time-delay induced instabilities, and the coupling between yaw and lateral dynamics. This literature survey synthesizes state-of-the-art PID steering control approaches for lane keeping and path tracking, covering (i) vehicle models used for controller design; (ii) PID control architectures (single-loop, cascaded/double-loop, feedforward + feedback, fuzzy/adaptive PID); (iii) tuning and optimization approaches (Ziegler–Nichols, fuzzy scheduling, genetic algorithms, PSO/MVO, minimum-variance adaptive PID); (iv) robustness and safety implications of delay and nonlinear behavior; and (v) validation practices spanning MATLAB/Simulink, CarSim co-simulation, ROS/Gazebo, hardware-in-the-loop (HIL), scaled test rigs, and real-vehicle experiments. The reviewed evidence indicates that while classical PID can be adequate for low-to-moderate curvature and low uncertainty, robust lane keeping in real-world conditions often requires hierarchical control, feed forward curvature compensation, gain scheduling/adaptation, and explicit attention to delay and nonlinear domains of attraction.

1. Context and Scope: Why PID Still Matters for Lane Keeping

Lane keeping is a foundational ADAS/AV function: it transforms perception outputs (lane boundaries or target path) into actuator commands (steering angle/torque). Even as advanced controllers (LQR/MPC/SMC) proliferate, PID remains prominent in industry practice and academic prototypes due to:

1. **Implementation simplicity** and low computation cost, favoring embedded ECUs and real-time constraints.
2. **Interpretability** for calibration engineers and safety assessors.
3. **Compatibility with hierarchical control** (outer-loop lateral control commanding inner-loop steering actuation).
4. **Ease of integration** with perception-driven error signals (cross-track error, heading error, yaw rate error).

However, literature consistently highlights that PID-only designs can be insufficient when speed varies, road curvature increases, disturbances occur, or delays become non-negligible. Several works treat PID as (i) baseline, (ii) a compensator around an optimal controller, or (iii) an adaptive/fuzzy scheduled structure [5], [7], [8], [10]–[15].

2. Lane Keeping Control Problem Definition

2.1 Control Objectives

A lane-keeping controller typically targets:

- **Lateral error minimization:** keep the vehicle near lane center (or reference path).
- **Heading/yaw alignment:** ensure vehicle orientation matches lane tangent.
- **Stability constraints:** avoid excessive sideslip/yaw rate.
- **Comfort constraints:** limit steering rate/jerk and oscillations.

2.2 Canonical Error Signals

The literature converges on two primary lateral control errors:

- **Cross-track (lateral) error** e_y : perpendicular distance to lane centerline.
- **Heading error** e_ψ : difference between vehicle yaw angle and lane tangent angle.

Many implementations use camera-based lane detection to estimate e_y and e_ψ (sometimes denoted y and Ψ) in a lane-fixed coordinate system [11], [12], [14]. Others use a dynamics error model with state vector including lateral velocity, yaw rate, or sideslip [10], [13], [15]

3. Modeling Foundations for PID Steering Control

3.1 Kinematic Bicycle Model (Low-to-Moderate Dynamics)

Kinematic models are common in early-stage lane keeping and student platforms. A typical kinematic bicycle form relates yaw rate to steering angle:

$$\dot{\psi} = \frac{v}{L} \tan \delta \approx \frac{v}{L} \delta$$

under small-angle assumption [11].

Key implication for PID: kinematic models are computationally cheap but can underrepresent high-speed behavior, tire saturation, and steering actuator effects. They can still support PID outer-loop designs when the inner loop and vehicle operate in a mild regime.

3.2 Dynamic Bicycle Model / 2-DOF Lateral Dynamics

For realistic lane keeping and stability, dynamic bicycle models appear repeatedly. A standard 2-DOF model uses lateral velocity v_y and yaw rate r , with linear tire cornering stiffness:

$$F_{yf} = C_f \alpha_f, F_{yr} = C_r \alpha_r$$

with slip angles typically:

$$\alpha_f \approx \delta - \beta - \frac{l_f r}{v}, \alpha_r \approx -\beta + \frac{l_r r}{v}$$

and state-space forms are derived for feedback design [10], [12], [13], [15]

3.3 Steering Actuator Dynamics, Saturation, and Delay (High Impact)

A major industry-relevant refinement is explicit inclusion of:

- **steering mechanism dynamics**
- **saturation (steering angle limits)**
- **feedback delay** (sensor + processing + actuation)

Vörös and Takács show that delay and steering dynamics can create unsafe regions *within linearly stable gains*, due to nonlinear limit cycles and subcritical Hopf bifurcations, validated experimentally on a laboratory rig [2]. This is crucial: **“stable in linear theory” does not guarantee safety under perturbations.**

4. PID Control Fundamentals for Steering and Lane Keeping

4.1 Continuous PID Law (General)

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

where $u(t)$ is steering command (angle or torque) and $e(t)$ is a chosen error metric (e.g., e_y , e_ψ , or a combination).

4.2 Lane-Keeping PID: Typical Error Definitions

A common structure is:

$$e(t) = w_y e_y(t) + w_\psi e_\psi(t)$$

and steering is:

$$\delta(t) = K_p e(t) + K_i \int e(t) dt + K_d \dot{e}(t)$$

This matches lane error definitions produced by camera-based lane models [11], [12], [14], and control comparisons in simulation [5], [7].

4.3 Incremental PID (Practical Discrete Implementation)

In embedded systems, incremental or discrete PID avoids integral windup and numerical issues:

$$\Delta u_k = K_p(e_k - e_{k-1}) + K_i e_k + K_d(e_k - 2e_{k-1} + e_{k-2})$$

(implementation form varies by discretization).

Industry relevance: discrete PID is easier to implement with saturations, anti-windup, rate limits, and safety monitors.

5. PID Steering Architectures in the Literature

5.1 Single-Loop PID Steering (Baseline)

Single-loop PID maps lateral deviation error to steering. This is frequently used as a baseline in comparative studies with MPC [5], [7] and is also adopted in ROS/Gazebo prototypes [12].

Observed limitations across studies:

- Increasing oscillations with speed or curvature [5], [12].
- Sensitivity to vehicle mass and cornering stiffness variation [5].
- Difficulty handling delay and nonlinearities robustly [2].

5.2 Cascaded/Double-Loop PID Structures (Industry-Realistic)

Törő et al. present a **double-loop control** approach for lane keeping on an autonomous go-kart. The camera outputs **lateral error** y and **yaw error** Ψ ; control uses an inner loop acting on yaw error and an outer loop acting on lateral error. This structure improves response by correcting yaw error before it integrates into large lateral deviation [11].

They also incorporate **feedforward disturbance compensation** for curvature, improving performance and stability margin, and tune controllers in HIL [11].

Key takeaway: double-loop control aligns with practical automotive steering control stacks (outer lateral controller + inner steering actuator controller).

5.3 Feedforward + Feedback: Curvature Compensation

Feedforward terms cancel known curvature-induced effects, treating curvature as a measurable “disturbance.” In Törő et al., curvature compensation is computed analytically using transfer functions [11]. In broader literature, feedforward is also paired with LQR/preview control; a notable design uses curvature preview feedforward + LQR feedback [10]. This architecture reduces corner-cutting and lag, especially on curves.

5.4 PID as a Compensation Layer around Optimal Controllers

A strong pattern in the reviewed works is **optimal controller + PID compensation**:

- A feedforward + predictive LQR is proposed with GA optimization of weights and a **PID steering compensation** term $\Delta\delta$ to correct lateral error and model mismatch [13].
- Hybrid MPC + PID frameworks appear where MPC handles lateral constrained optimization while PID handles speed or compensates modeling uncertainty [8].

Industry interpretation: PID often serves as a “practical stabilizer” where model-based controllers may be sensitive to mismatch.

6. Tuning and Adaptation of PID for Lane Keeping

6.1 Manual/Heuristic Tuning and Ziegler–Nichols

Ziegler–Nichols and manual tuning remain common baselines, but are not robust to varying speeds and road conditions [5], [7], [12].

6.2 Fuzzy Gain Scheduling (Fuzzy-PID)

Fuzzy-PID appears repeatedly as a response to parameter variation and poor damping:

- Fuzzy-PID for lane recognition and lane keeping shows reduced oscillations and improved settling vs PD/MPC baselines in simulation [7].
- ROS/Gazebo implementation uses fuzzy logic to tune K_p , K_i , K_d based on error and error derivative; results report lower overshoot and faster settling than conventional PID [12].
- Earlier vehicle stability work shows fuzzy logic has better robustness while PID is simpler to deploy [1].

Representative fuzzy-PID formulation [12]:

$$\psi_{\text{tuning}}(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de}{dt}$$

with fuzzy inference adjusting gains based on $e, \Delta e$.

Strengths: improved adaptability without full model dependence.

Risks: rule design and membership functions can be subjective; verification burden increases.

6.3 Adaptive PID via Minimum Variance / Online Parameter Adjustment

Zhao et al. present an adaptive PID approach for the “Intelligent Pioneer” vehicle, motivated by the inability of fixed PID gains to guarantee performance under parameter changes and disturbances. Their adaptive PID is grounded in generalized minimum variance control, updating parameters online to reduce output variance and maintain low tracking error during turns [15]. Real-world competition performance provides strong validation context.

Industry value: online adaptation improves robustness across changing conditions; however, requires careful safety bounding.

6.4 Metaheuristic Optimization (GA/PSO/MVO) for PID Parameters

Optimization methods (GA/PSO/MVO) are used widely to tune control parameters:

- GA is used to optimize LQR weights, and PID provides steering compensation [13].
- PSO-PID is used for longitudinal control coordinated with lateral MPC; dynamic PSO parameters improve search performance [4].
- PSO/MVO comparisons for EV speed show optimization reduces overshoot and steady-state error, demonstrating the general value of metaheuristic tuning for PID families [6].

Engineering interpretation: optimization can reduce calibration time and improve objective metrics, but must be validated against unseen operating regimes.

7. Robustness and Safety: What Breaks PID Lane Keeping

7.1 Parameter Variations (Mass, Cornering Stiffness, Speed)

Simulation studies show classical PID/PD lane keeping is less robust when:

- vehicle load (mass) increases,
- cornering stiffness changes (road conditions),
- longitudinal speed increases

MPC tends to be more robust under these variations because of prediction and constraint handling [5]. Fuzzy-PID tends to be more stable than conventional PID under speed variation [7], [12].

7.2 Time Delay and Nonlinear Instabilities

A critical contribution to safety understanding is the nonlinear delay analysis in lane keeping [2]. The authors show that:

- time delay shrinks stable gain regions,
- subcritical Hopf bifurcations can produce **unstable limit cycles** around a linearly stable equilibrium,
- hence **unsafe regions exist inside stable parameter maps**.

This is highly relevant to PID-like tuning practices: purely linear tuning may yield gains that look stable but produce unsafe dynamics under disturbances.

7.3 Actuator Saturation and Steering Dynamics

Steering saturation and actuator dynamics materially change stability regions [2]. Controllers must consider:

- steering rate/angle constraints,
- inner-loop torque/servo control,
- saturation smoothing (for analysis) and anti-windup (for implementation).

7.4 Stability Variables: Yaw Rate and Sideslip

Vehicle stability control literature emphasizes yaw rate and sideslip as critical variables; PID and fuzzy controllers can regulate yaw/sideslip and improved combined strategies can outperform isolated ones [1]. Although not

lane-keeping-specific, this informs safe steering control design because excessive yaw/sideslip can precede lane departure and loss of control.

8. PID vs MPC vs LQR (Positioning PID in the Control Stack)

8.1 Comparative Findings

Across comparisons:

- MPC typically achieves better tracking on complex paths but is computationally heavier [5], [8].
- PID/PD is simpler and can work well on mild paths but can oscillate and degrade under parameter changes [5], [12].
- LQR provides optimal feedback for linearized models but can face steady-state error or lag unless feedforward/prediction is added [10], [13].

8.2 Practical Hybridization Trend

Modern designs frequently combine:

- PID for speed (longitudinal) + MPC/LQR for steering (lateral) [4], [8], [10].
- feedforward preview + feedback optimal control + PID compensation [10], [13].

This reflects industry practice: keep the real-time stack reliable and computationally tractable while adding higher-level optimization where justified.

9. Validation Practices: From Simulation to Industry-Grade Evidence

9.1 MATLAB/Simulink and CarSim Co-Simulation

Co-simulation enables high-fidelity vehicle dynamics evaluation while prototyping control logic [4], [8], [13].

9.2 Hardware-in-the-Loop (HIL)

HIL is highlighted as essential for safe iteration, reproducibility, and development speed; Törő et al. used HIL for controller tuning and validation prior to deployment [11].

9.3 ROS/Gazebo Prototyping

ROS/Gazebo allows integrated perception-control simulation. Fuzzy-PID lane keeping in ROS/Gazebo demonstrates improved overshoot/settling vs PID [12].

9.4 Experiments and Competition Validation

Real vehicle experiments and competitions provide strong evidence:

- Adaptive PID in a competition vehicle (“Intelligent Pioneer”) achieved <0.5 m tracking error and robust speed control under real conditions [15].
- Delay and nonlinear experimental validation on scaled rigs demonstrates unsafe gain zones [2].

10. Consolidated Evidence Tables

Table 1 — Summary of PID-Relevant Literature for Lane Keeping and Steering Control

Ref.	Primary Focus	Vehicle Model	Control Strategy	Validation	Key Takeaway
[11]	Lane keeping go-kart	Kinematic bicycle	Double-loop + feedforward; Stanley	HIL + CarSim/Simulink	Cascaded loops + feedforward reduce errors; HIL essential
[12]	Lane keeping in ROS	Bicycle model (linear steering)	Fuzzy-PID steering	Gazebo/ROS	Fuzzy scheduling reduces overshoot & settling time
[5]	Lane keeping comparison	Bicycle model	PD/PID vs MPC	MATLAB/Simulink	PID/PD less robust to mass/speed/stiffness than MPC

[7]	Lane recognition + lane keeping	Bicycle model	Fuzzy-PID vs PD vs MPC	Simulation	Fuzzy-PID improves settling and reduces oscillation
[2]	Delay + nonlinear safety	Single track + steering dynamics	Linear feedback w delay	Numerical bifurcation + lab rig	Unsafe zones exist inside linear stability due to limit cycles
[15]	Adaptive PID vehicle	2-DOF dynamic model	Adaptive PID (minimum variance)	Real vehicle + competition	Online adaptation improves robustness under disturbances
[13]	Predictive LQR + PID compensation	Tracking error model	FF + predictive LQR + PID	CarSim/Simulink	GA tuning + PID compensation improves accuracy & stability
[10]	Hybrid steering & speed	Dynamic bicycle	Curvature-based speed + PID + preview + LQR	Simulation + experiments	Curvature-aware speed + combined steering reduces curve errors

[1]	Vehicle stability control	2-DOF + yaw moment	PID vs fuzzy; combined	Simulation	Fuzzy robust; PID practical; combined strategies best
[4]	Coordinate d long+lat	Nonlinear long + LPV lateral	PSO-PID speed + LPV-MPC steer	CarSim co-sim	Optimization improves tracking and respects constraints

11. Figures

Fig. 1 — Hierarchical Lane Keeping Control Stack

AUTONOMOUS VEHICLE LANE KEEPING ARCHITECTURE

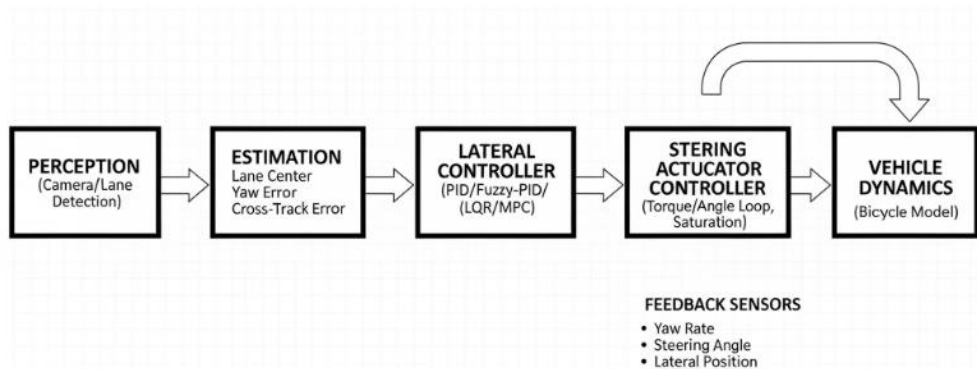


Fig. 2 — Cross-Track Error and Heading Error Geometry

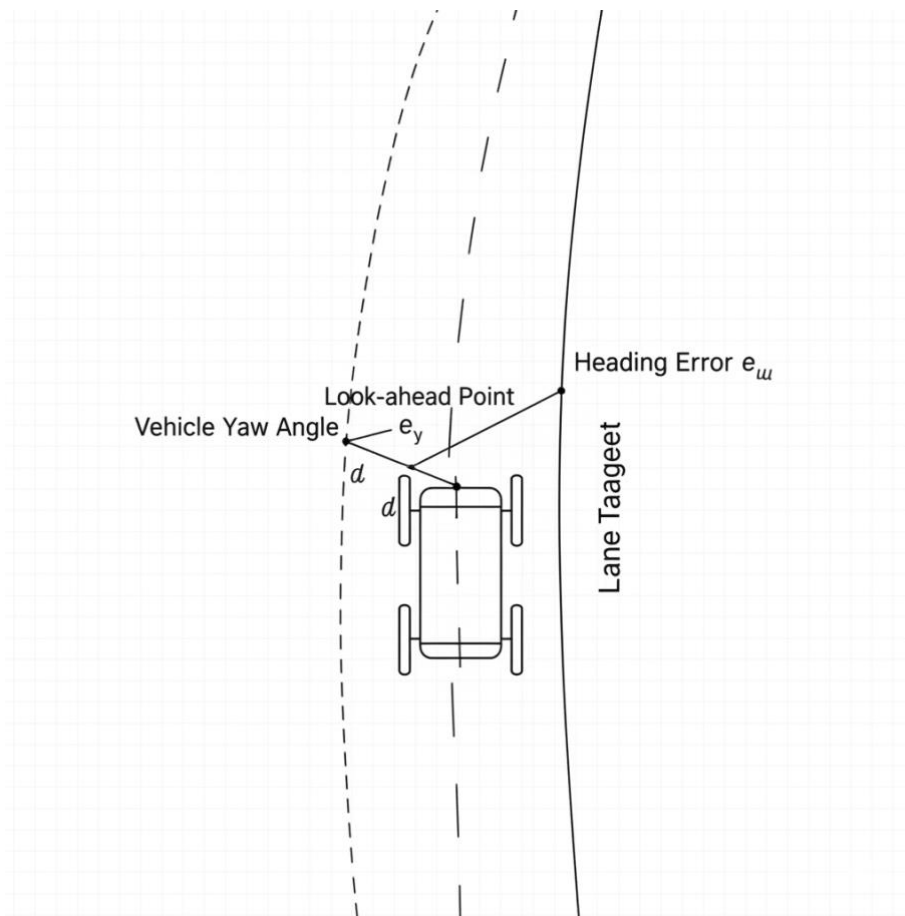


Fig. 3 — PID vs Cascaded (Double-loop) Steering Control

COMPARISON OF STEERING CONTROL SYSTEMS FOR VEHICLE TRAJECTORY TRACKING

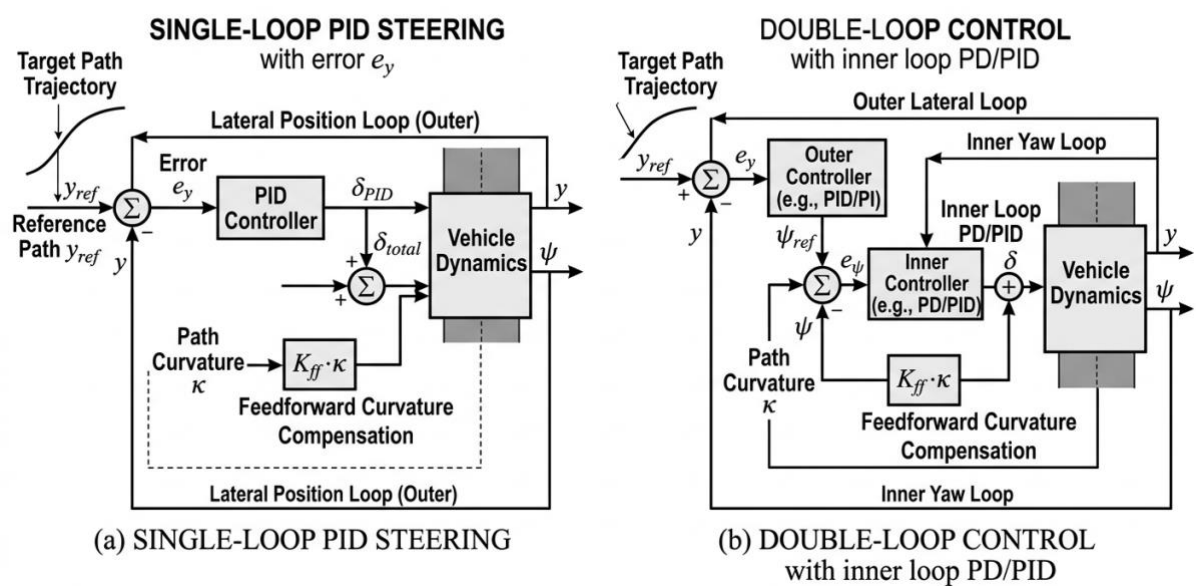


Fig. 4 — Fuzzy-PID Membership Functions and Rule Base

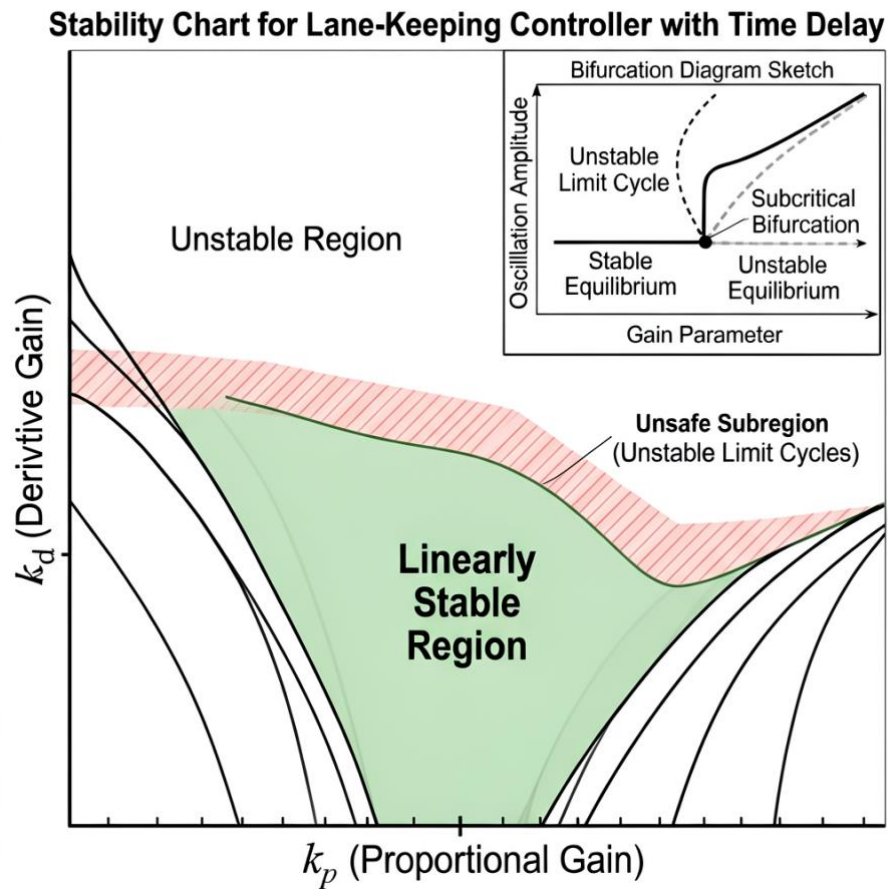


Fig. 5 — Delay-Induced Unsafe Zones (Limit Cycles)

FUZZY LOGIC PID SELF-TUNING SYSTEM CONFIGURATION

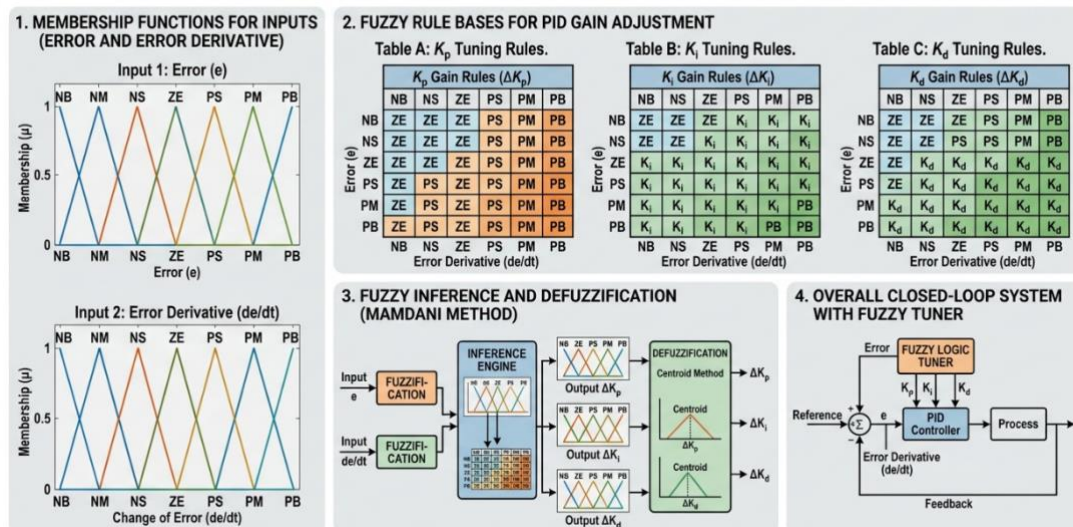
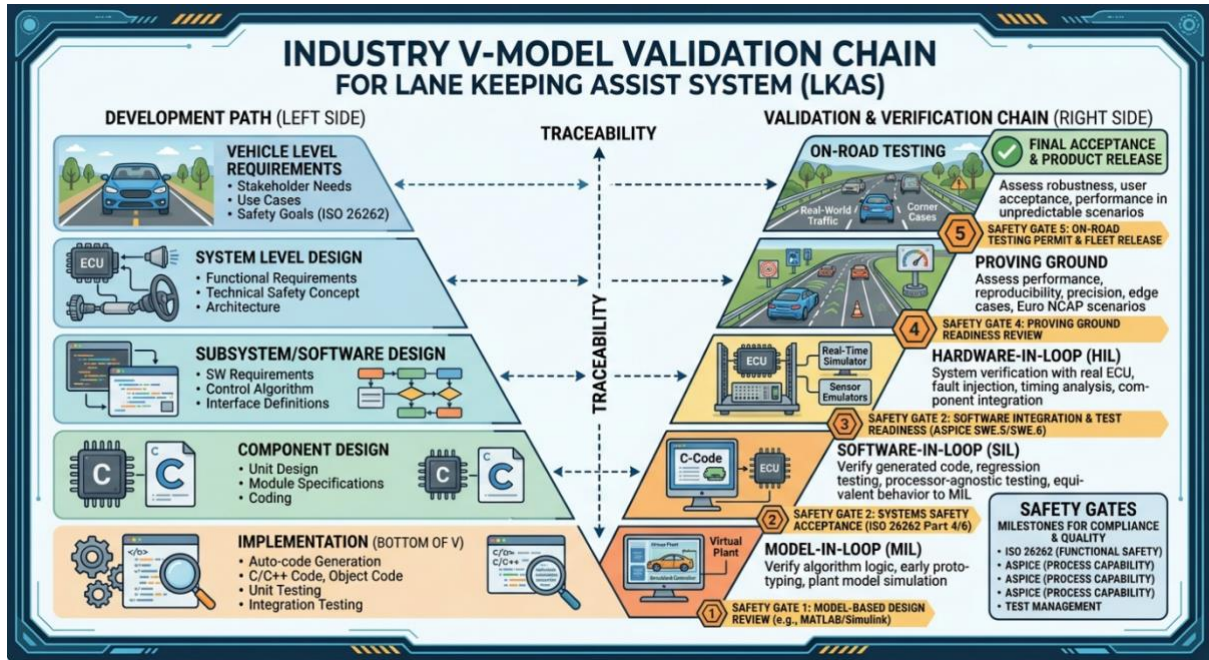


Fig. 6 — Validation Pipeline (MIL → SIL → HIL → Vehicle)



12. Key Equations Set (Insert as “Mathematical Foundations”)

12.1 Error Signals

$$e_y(t) = y_{\text{ref}}(t) - y(t), e_\psi(t) = \psi_{\text{ref}}(t) - \psi(t)$$

12.2 PID Steering

$$\delta(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

with $e(t) = w_y e_y(t) + w_\psi e_\psi(t)$.

12.3 Curvature Feedforward (Example Form)

Use a curvature-dependent steering feedforward:

$$\delta_{ff}(t) = L \cdot \kappa(t)$$

where κ is path curvature; refined versions appear in preview/feedforward designs [10], [11].

12.4 Steering Compensation (PID Add-on)

$$\delta(t) = \delta_{\text{base}}(t) + \Delta\delta(t), \Delta\delta(t) = a_1 e_y + a_2 \int e_y dt + a_3 \frac{de_y}{dt}$$

consistent with PID compensation layered on optimal control [13].

13. Research Gaps

1. **Delay-aware PID tuning is underreported** in many lane-keeping implementations, despite strong evidence that delay reshapes stability and safety regions [2]. Many PID papers assume negligible delay or treat it implicitly.
2. **Nonlinear safety analysis is rarely integrated** into controller calibration workflows. The presence of unstable limit cycles implies a need for “safe tuning zones” rather than “best linear damping points” [2].
3. **Robustness reporting is inconsistent**: many works report only average tracking error, but omit disturbance profiles (wind, friction, sensor noise), actuator saturation events, or uncertainty bounds.
4. **Perception–control coupling is frequently simplified**: lane detection failures, intermittent markings, and occlusions can produce discontinuous errors. Controllers should be evaluated with perception dropouts and noise models, as discussed in lane detection surveys and vision-based pipelines [9], [11], [12].
5. **Industry validation expectations** increasingly require multi-stage evidence (MIL/SIL/HIL/vehicle) and safety argumentation; HIL is proven beneficial but not universally adopted across academic PID studies [11].

14 Science Effects

PID-based lane keeping in autonomous vehicles is influenced by multiple scientific effects arising from real-world conditions and system limitations. One major effect is sensor noise, where camera-based lane detection may produce inaccurate cross-track and heading errors due to poor lighting, weather conditions, or occlusions. This directly impacts controller performance.

Another important effect is tire–road interaction, which introduces nonlinear behavior such as slip and saturation. At higher speeds or during sharp turns, the relationship between steering input and vehicle response becomes nonlinear, reducing the effectiveness of simple PID control.

Time delay is a critical factor. Delays from sensing, computation, and actuation can destabilize the system, causing oscillations or even unsafe behavior. Research shows that even when a system appears stable in theory, delays can introduce nonlinear instabilities like limit cycles.

Additionally, parameter variation effects occur due to changes in vehicle mass, speed, and road conditions. These variations affect system dynamics and can make fixed PID gains ineffective.

Finally, computational effects influence real-time performance. While PID is computationally efficient, advanced enhancements (adaptive, fuzzy, or hybrid control) increase processing requirements, which must still meet real-time constraints.

15 Concepts, Rules, Principles, Laws

15.1 Concepts

The core concept is lane keeping control, where the vehicle continuously adjusts its steering to remain within lane boundaries. This is achieved by minimizing cross-track error (lateral deviation from the lane center) and heading error (difference in orientation).

The system relies on PID control, which uses proportional, integral, and derivative components to correct errors. More advanced concepts include cascaded (double-loop) control, where one loop controls lateral position and another controls yaw, improving stability and response.

Another important concept is feedforward + feedback control, where feedforward compensates for known factors like road curvature, while feedback corrects real-time errors. The system also depends on vehicle models (kinematic and dynamic bicycle models) to represent motion behavior.

15.2 Rules

Lane-keeping systems follow a structured control rule:

- Measure error (cross-track and heading)
- Apply control using PID
- Generate steering command

- Vehicle responds and updates position

This forms a continuous loop.

Additional rules include:

- **PID tuning rules** (e.g., Ziegler–Nichols, optimization methods)
- **Data flow rules** in cascaded systems (outer loop → inner loop)
- **Constraint rules** such as steering limits and actuator saturation

These rules ensure stable and realistic system behavior.

15.3 Principles

The system is based on several key control principles:

- **Feedback principle:** The controller continuously adjusts steering based on error signals.
- **Error minimization principle:** The objective is to reduce both lateral and orientation errors.
- **Stability principle:** The system must avoid oscillations and divergence.
- **Adaptation principle:** Controller parameters may need to change based on speed or environment.
- **Robustness principle:** The system should perform well despite disturbances and uncertainties.

These principles guide the design of reliable lane-keeping systems.

15.4 Laws

The system operates under fundamental physical and control laws:

- **Vehicle motion laws:** Govern how steering affects movement (kinematics and dynamics).
- **Control system stability laws:** Define conditions for stable response.
- **Nonlinear system behavior laws:** Real systems exhibit nonlinearities due to tire dynamics and actuator limits.
- **Delay-induced instability laws:** Time delays can shift stable systems into unstable regions.

These laws highlight that real-world performance often differs from ideal theoretical models.

16 Constants and Variables

16.1 Constants

Constants are fixed parameters that define the system and do not change during operation. These include:

- PID gains (K_p, K_i, K_d) when not adaptive
- Vehicle parameters such as wheelbase, mass, and tire stiffness
- Sampling time used in digital control systems
- Physical constraints like maximum steering angle

These constants determine how the controller behaves under normal conditions.

16.2 Variables

Variables are dynamic values that continuously change during operation. These include:

- Cross-track error (lateral deviation)
- Heading error (orientation difference)
- Steering angle (control output)
- Vehicle speed and yaw rate
- Sensor measurements from cameras or IMU

These variables interact with each other in real time. For example, changes in steering angle affect vehicle motion, which in turn updates the error values. The controller continuously adjusts based on these changing variables to maintain lane position.

13 Equations & Relationships

1. Cross-Track Error

$$e_y = y_{vehicle} - y_{lane}$$

Relationship:

Difference between vehicle position and lane center → main input to controller.

2. Heading Error

$$e_\psi = \psi_{vehicle} - \psi_{lane}$$

Relationship:

Represents orientation mismatch → affects turning direction.

3. PID Control Law

$$\delta(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$$

Relationship:

- K_p : reacts to current error
- K_i : removes steady-state error
- K_d : predicts future error and reduces oscillation

4. Combined Error Model

$$e(t) = e_y + \lambda e_\psi$$

Relationship:

Total error combines position and orientation → improves control accuracy.

5. Kinematic Bicycle Model

$$\dot{\psi} = \frac{v}{L} \tan(\delta)$$

Relationship:

Steering angle controls yaw rate → determines vehicle turning.

6. Curvature Feedforward

$$\delta_{ff} = L \cdot \kappa$$

Relationship:

Road curvature directly affects steering → reduces tracking error in curves.

System of Equations

The lane-keeping system is governed by a combined set of equations:

1. Error equations:

$$e_y, e_\psi$$

2. Control equation:

$$\delta(t) = K_p e + K_i \int e dt + K_d \frac{de}{dt}$$

3. Vehicle dynamics:

$$\dot{\psi} = \frac{v}{L} \tan(\delta)$$

4. Feedforward + feedback:

$$\delta = \delta_{ff} + \delta_{PID}$$

Relationship:

- Errors $\rightarrow$ input to controller
- Controller $\rightarrow$ generates steering
- Vehicle dynamics $\rightarrow$ produce motion
- Motion $\rightarrow$ updates errors

This forms a closed-loop system.

14 What-if Analysis

The behavior of the lane-keeping system can change significantly under different scenarios:

- **What if vehicle speed increases?**
Higher speed amplifies system sensitivity, leading to oscillations and reduced stability if PID gains are not adjusted.
- **What if time delay increases?**
Delays can cause instability and oscillatory behavior, even if the system is theoretically stable.
- **What if road curvature increases?**
Simple PID control may struggle to track sharp curves accurately, requiring feedforward compensation or advanced control methods.
- **What if sensor noise or failure occurs?**
Incorrect error measurements can lead to improper steering commands and degraded performance.
- **What if vehicle parameters change (mass, stiffness)?**
Fixed PID gains may not adapt well, reducing accuracy and stability.

- **What if actuator saturation occurs?**

Steering limits can prevent the controller from applying the required correction, causing tracking errors.

This analysis shows the limitations of basic PID and the need for adaptive or hybrid approaches.

15 Tools / Technologies / Methods

Various tools and methods are used to design, test, and improve PID-based lane-keeping systems:

- **Control Methods**
 - Classical PID control
 - Fuzzy-PID for adaptive gain tuning
 - Adaptive PID for real-time parameter adjustment
 - Hybrid approaches combining PID with MPC or LQR
- **Optimization Techniques**
 - Genetic Algorithms (GA)
 - Particle Swarm Optimization (PSO)
 - Multi-Verse Optimization (MVO)

These methods help in tuning controller parameters effectively.

- **Simulation Tools**
 - MATLAB/Simulink for control design and analysis
 - CarSim for realistic vehicle dynamics simulation
 - ROS/Gazebo for integrated robotics simulation
- **Validation Methods**
 - Model-in-the-Loop (MIL)
 - Software-in-the-Loop (SIL)
 - Hardware-in-the-Loop (HIL)
 - Real vehicle testing

These stages ensure that the controller works correctly before real-world deployment.

- **Technologies**

- Camera-based lane detection systems
- IMU sensors for motion estimation
- Embedded systems (ECUs) for real-time control
- Communication systems for integration with other vehicle modules

16 Conclusion

This literature survey underscores the enduring relevance of PID-based steering control in autonomous vehicle lane-keeping, while also highlighting its limitations under real-world operating conditions. Classical PID controllers remain attractive due to their simplicity, transparency, and ease of deployment on embedded platforms. However, the reviewed evidence makes clear that fixed-gain PID designs are vulnerable to speed variations, nonlinear tire dynamics, actuator saturation, and time delays, which can compromise both stability and safety.

To address these challenges, researchers and practitioners increasingly adopt enhanced PID architectures—such as cascaded double-loop structures, feedforward curvature compensation, fuzzy gain scheduling, adaptive minimum-variance tuning, and metaheuristic optimization. These approaches improve robustness, reduce oscillations, and extend the domain of safe operation, particularly when integrated into hierarchical control stacks or hybrid frameworks alongside MPC and LQR.

Validation practices further emphasize the importance of multi-stage testing, from MATLAB/Simulink and CarSim co-simulation to hardware-in-the-loop and real-vehicle experiments. Such rigorous pipelines ensure that controllers not only perform well in theory but also withstand uncertainties in perception, dynamics, and computation.

In summary, PID control continues to serve as both a practical baseline and a versatile stabilizing layer in autonomous lane-keeping systems. Its future role lies not in isolation, but in synergy with adaptive, feedforward, and model-based strategies—anchored by delay-aware tuning and nonlinear safety analysis. This integrated perspective ensures that PID remains a cornerstone of reliable, industry-ready autonomous steering solutions.

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